

UNIFOUND MANAGEMENT SYSTEM

COMPLETE APPLICATION USER MANUAL & OPERATING GUIDE

A Step-by-Step Practical Guide for Students, Faculty & Campus Administrators



SYSTEM ACCESS QUICK REFERENCE

- 🌐 **Web Application:** <https://unifound-app.vercel.app>
- ⚡ **Backend API & Swagger:** <https://unifound-mqga.onrender.com>
- 🔑 **Default Admin ID:** admin@campus.edu
- 🔑 **Default Password:** AdminPass123!

Project Team Members:

Sahil Khot (2303051240190) • Shivani Choudhary (2303051240212)
Minal Prasad (2303051240122) • Shivang Bhadwal (2303051240211)

Project Guide: Mrs. Arpita Meet Vaidya

Table of Contents & Quick Overview

1. Introduction to UniFound System – Purpose, Core Features, Architecture	Page 3
2. Account Registration & Authentication – Sign Up, Login, Profile Management	Page 4
3. Reporting Lost & Found Items – Categories, Locations, Photo Uploads	Page 5
4. Browsing, Searching & AI Matching – Filters, AI Confidence Engine	Page 6
5. Submitting & Managing Claims – Proof of Ownership, Claim Lifecycle	Page 7
6. Administrator Portal & Moderation – Claim Approval, Item Status, Audit Logs	Page 8
7. Advanced Capabilities (MCP Agent & Vision) – Automated Discovery & AI Chat	Page 9
8. Troubleshooting & Frequently Asked Questions (FAQ) – Security & Help Desk	Page 10

System Architecture & User Roles

UniFound operates with two distinct privilege tiers to guarantee secure handover and fraud prevention:

**Campus User (Student / Faculty)**

- Report misplaced or recovered articles with images
- Search active listings via keywords & categories
- Receive automated AI match notifications
- File claim requests with private proof of ownership
- Track real-time status of claims and personal reports

**Campus Administrator (Security Office)**

- Review pending ownership claims and submitted proofs
- Approve or reject claims with official comments
- Manage item lifecycle (Open → Claim Pending → Claimed)
- Oversee campus security custody of high-value items
- Access comprehensive audit trail for institutional accountability

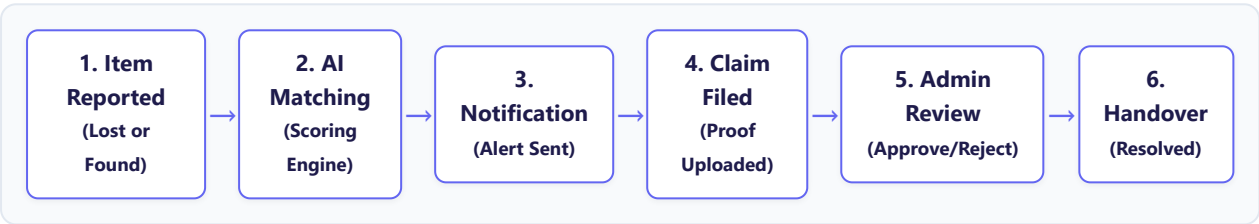
 **Cloud Deployment Status:** The application is fully live and cloud-hosted. Front-end assets are served globally via **Vercel CDN** (<https://unifound-app.vercel.app>), while backend REST endpoints execute on **Render Cloud** (<https://unifound-mqga.onrender.com>) backed by PostgreSQL database and JWT token security.

1. Introduction & Core Operating Workflow

UniFound is an enterprise-grade campus Lost & Found management portal developed specifically for university environments. Traditional campus paper registers and unorganized bulletin boards suffer from high recovery latency, loss of claimant privacy, and vulnerability to fraudulent claims. UniFound solves these challenges through structured digital reporting, automated multi-factor AI matching, and centralized administrative verification.

The Complete End-to-End Recovery Lifecycle

The diagram below illustrates the operational pathway taken from the moment an item is misplaced or found on campus until it is safely reunited with its rightful owner:



Core Technological Highlights

 **AI Multi-Factor Matching**

Whenever an item is submitted, the system automatically runs a weighted matching algorithm evaluating: Category match (35%), Text & keyword similarity (30%), Date proximity (20%), and Visual color/metadata (15%). If the confidence score exceeds 60%, instant alerts are created.

 **Secure Verification & Privacy**

Identifying marks, serial numbers, and private item attributes are never published publicly. Only administrators can review claimant proofs, preventing unauthorized individuals from falsely claiming valuables.

 **Automated Image Analysis**

Integrated Pillow (PIL) computer vision extracts dominant color profiles, aspect ratios, and format attributes to enrich item listings and assist automated visual comparisons.

 **Model Context Protocol (MCP)**

UniFound includes an integrated MCP tool server that enables LLM agents to autonomously query lost/found records, perform natural language diagnostics, and assist users conversationally.

 **Accessing the System:** Open any modern web browser (Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari on iOS/Android) and navigate to <https://unifound-app.vercel.app>.

2. Account Registration, Login & Profiles

To ensure security and maintain a verifiable chain of custody, reporting items and filing claims requires an authenticated account.



Step 1: Registering a New Student / Faculty Account

- 1 Navigate to the Registration Page:** Click **"Register"** on the top navigation bar or browse to `https://unifound-app.vercel.app/register`.
- 2 Fill in Required Information:**
 - **Full Name:** Enter your official university name (e.g., *Minal Prasad*).
 - **Email Address:** Provide your active university or personal email (e.g., *minal@paruluniversity.ac.in*).
 - **Password:** Choose a strong password (minimum 8 characters with letters, numbers, and symbols).
 - **Phone Number (Optional):** Mobile number for SMS/call verification.
 - **Department:** Department name (e.g., *Computer Science & Engineering*).
 - **Student / Staff ID:** University Enrollment Number (e.g., *2303051240122*).
- 3 Submit:** Click **"Create Account"**. Your account is immediately activated with the standard **USER** role.



Step 2: Logging In

- 1** Go to the Login Page at `/login`.
- 2** Enter your registered **Email Address** and **Password**.
- 3** Click **"Sign In"**. Upon successful authentication, the server generates an encrypted JSON Web Token (JWT) with 24-hour validity stored securely in client storage.



Managing Your Profile

Click on your avatar or username at the top right corner and select **"Profile"** (`/profile`) to:

- Update contact numbers and department details.
- Review your personal activity statistics (Total items reported, Active claims, Resolved recoveries).
- View your recent match alerts and security status.



Password Security Best Practice: Never share your login credentials with others. If you suspect your account has been compromised, contact the security department immediately.

3. How to Report Lost & Found Items

Whether you have misplaced personal property or found an unattended article on campus, reporting it promptly maximizes the recovery rate.

 **Reporting a LOST Item**

Use when you have lost an item on campus.

URL: /report-lost

 **Reporting a FOUND Item**

Use when you have discovered an unattended item.

URL: /report-found

Field-by-Field Submission Guide

Field Name	Type / Requirement	Guidance & Best Practices
Item Title	Text (Required)	Be clear and specific (e.g., "Black Dell G15 Laptop Charger" instead of just "Charger").
Item Type	Selection (Required)	Select LOST or FOUND .
Category	Dropdown (Required)	Choose from: <i>Electronics, Identity & Documents, Keys, Wallets & Bags, Clothing, Books, Other.</i>
Campus Location	Text / Select (Required)	Specify exact building & room (e.g., "Central Library 2nd Floor, Reading Room B").
Date & Time	Date Picker (Required)	Estimated date and time when the item was lost or discovered.
Description	Textarea (Required)	Provide physical traits: color, brand, stickers, scratches, wear marks, case type.
Photograph	Image Upload (Recommended)	Upload a clear JPG/PNG photo. Helps the AI matching engine and visual verification.
Custody / Handover	Text (Found items only)	Where the item is currently kept (e.g., "Handed over to PIT Security Desk Main Gate").

 **Critical Privacy Rule for High-Value Items:** For items containing sensitive data or high monetary value (such as wallets with cash or laptops with passwords), do **NOT** post exact PINs, passwords, or total cash amounts in the public description. Keep those details as private proof for the claim verification stage!

4. Search System & Automated AI Matching

How to Search Campus Items

The Search Portal (<https://unifound-app.vercel.app/search>) provides instant query filtering across all publicly active lost and found listings:

- **Keyword Search:** Type keywords such as brand names ("Apple", "Sony", "Titan"), colors, or descriptions.
- **Category Filter:** Filter exclusively by Electronics, Keys, Wallets, Documents, etc.
- **Status Toggle:** Switch between *Active/Open* items and *Resolved* items.
- **Location Filter:** Narrow down to specific campuses, buildings, or cafeterias.

The Automated AI Matching Engine

UniFound features a background intelligence pipeline that continuously evaluates newly reported items against existing database records to detect potential matches automatically.

Matching Factor	Weight	How It Works
Category Match	35%	Ensures matching items belong to the same functional taxonomy (e.g., Electronics to Electronics).
Text Similarity	30%	Analyzes title keywords, token intersections, and description phrases using fuzzy text scoring.
Date Proximity	20%	Calculates temporal difference between loss date and discovery date (highest score if within 48h).
Visual / Location	15%	Compares campus zone proximity and automated Pillow visual characteristics (dominant color profiles).

Understanding Confidence Score Badges

 **High Match (80% – 100%)**
Strong match on category, title, date, and location.

 **Moderate Match (60% – 79%)**
Partial overlap in description or neighboring date.

 **Low Match (< 60%)**
Weak correlation; manual review advised.

 **Instant Match Notifications:** When a match score exceeds 60%, UniFound automatically generates an alert in your notification center (</notifications>) with a direct link to the matched item so you can immediately submit an ownership claim!

5. How to File an Ownership Claim

When you discover an item in the public catalog that belongs to you, you must submit a formal **Claim Request** to establish genuine ownership before physical handover can take place.

Step-by-Step Claim Submission Flow

- 1 Open Item Details:** Click on the item card from the search results or notification alert to view its details page (`/items/:id`).
- 2 Click "Claim This Item":** If the item is marked **FOUND** and its status is **OPEN**, the "Claim Item" button will be active.
- 3 Fill the Claim Form:**
 - **Detailed Proof of Ownership:** State unmentioned identifying details (e.g., lock-screen wallpaper, unique scratch under the charging port, engraved initials, serial number, contents of internal pouch).
 - **Supporting Proof / File Attachment:** Upload purchase invoices, photos of you using the item previously, or an ID card copy.
 - **Contact Number:** A phone number where the security office can reach you for handover.
- 4 Submit Claim:** Confirm the declaration and click **"Submit Claim"**.

The Claim Status Lifecycle

Track the progress of your submitted claims at `/my-claims` :

PENDING – Your claim has been received and queued for administrative review by campus security.

APPROVED – Your proof has been authenticated! You may proceed to the designated security desk with your university ID card to collect the item.

REJECTED – The provided proof did not match the physical characteristics. Admin notes will indicate the reason.

 **Anti-Fraud Institutional Policy:** Submitting fraudulent claims for articles belonging to others is a violation of the Parul University Student Code of Conduct and will result in disciplinary action and account deactivation.

6. Campus Administrator Portal & Moderation

The Administrator Portal is reserved for authorized campus security personnel and department coordinators to moderate reports, verify claims, and maintain campus-wide chain of custody.

Logging In as Administrator

Login URL: `https://unifound-app.vercel.app/login`

Default Admin ID: `admin@campus.edu`

Default Password: `AdminPass123!`

Admin Dashboard Features (/admin)

Real-time Metric Cards

Instantly monitor Total Items Reported, Active Lost Listings, Recovered Found Items, Pending Verification Claims, and Resolution Rates.

Item Catalog Moderation

Filter items across all categories, edit custody locations, archive stale entries, or mark items as Claimed / Resolved.

Processing Claim Requests (/admin/claims)

The Claim Moderation Queue displays all claims filed by campus users:

- 1 Review Claimant Information:** Inspect claimant's full name, student ID, department, and telephone number.
- 2 Inspect Submitted Ownership Proof:** Compare the claimant's private statement against the physical item in custody.
- 3 Take Moderation Action:**
 - **Approve Claim:** Marks the claim as **APPROVED** and transitions the item to **CLAIMED**. An automated notification instructs the student to collect the item.
 - **Reject Claim:** Provide an administrative comment explaining why the proof was insufficient (e.g., "Serial number mismatch").

 **Security Audit Trail:** Every administrative action (claim approval, rejection, item status modification) is logged with timestamp, administrator ID, and action description in the immutable `audit_logs` database table for institutional auditability.

7. AI Agent & Model Context Protocol (MCP)

UniFound integrates the open **Model Context Protocol (MCP)** standard, exposing tools that enable autonomous AI agents (such as Google Gemini, Claude, or local ReAct agents) to interact with the portal via natural language.

Built-In MCP Tools Reference

MCP Tool Name	Parameters	Functional Capability
<code>search_items</code>	<code>query, item_type, category</code>	Executes deep semantic search across all public lost & found records.
<code>get_item_details</code>	<code>item_id</code>	Retrieves complete metadata, custody location, and status for an item.
<code>analyze_image</code>	<code>image_url / file_path</code>	Executes computer vision analysis extracting color palette and dimensions.
<code>check_matches</code>	<code>item_id</code>	Invokes the multi-factor AI matching pipeline to find cross-pairings.

Using the Natural Language Agent Endpoint

Developers and advanced users can communicate directly with the UniFound Agent endpoint at `POST /api/v1/agent/query` :

```
# Sample Agent Request Payload
POST https://unifound-mqga.onrender.com/api/v1/agent/query
Content-Type: application/json

{{
  "prompt": "Did anyone find a brown leather Titan wallet near the Central Library yesterday?",
  "session_id": "user-session-101"
}}

# Agent Autonomous Execution:
1. Agent calls `search_items(query='Titan wallet', category='Wallets')`
2. Finds 1 item: ID #14 (Found at Library 1st floor desk)
3. Returns natural language summary with item ID, custody status, and link.
```

 **Swagger Interactive Testing:** Try the MCP tools directly in the interactive Swagger API sandbox at <https://unifound-mqga.onrender.com/docs> under the "Agent Direct" section.

8. FAQ & Troubleshooting Guide

? Frequently Asked Questions

Q1: What should I do immediately after losing an item?

A: Post a **LOST** report right away with specific physical traits and the estimated loss location. Regularly check your notifications for AI match alerts.

Q2: I found an item on campus. Should I keep it or hand it over?

A: Report it on UniFound with a photo, then immediately hand the article over to the nearest departmental security desk or Central Security Office. Record the custody location in your report.

Q3: How long does claim approval take?

A: Most claims are reviewed by campus security within 24 hours of submission. High-value electronics requiring serial number inspection may take up to 48 hours.

Q4: Why was my claim rejected?

A: Claims are rejected if the provided description or ownership proof contradicts the physical article. You may resubmit with additional documentation or visit the security desk in person.

🔧 Troubleshooting Common Issues

Issue	Solution
Cannot log in / Invalid credentials	Verify your email and password casing. If forgotten, contact campus security or use password reset.
Photo upload fails	Ensure your image file is in JPG, PNG, or WEBP format and does not exceed 10 MB.
Session expired alert	Security tokens expire after 24 hours of inactivity. Simply re-login at <code>/login</code> .
Admin Dashboard not visible	Admin features are restricted to accounts with the ADMIN role. Log in with the official administrator credentials.

📞 Campus Help Desk & Support Contact

Parul University Central Security & Student Affairs • Vadodara, Gujarat

Email: support@unifound-campus.edu • Portal: <https://unifound-app.vercel.app>